

Fundamental concept of boolean algebra

Only two mathematical operations

Addition

↑
OR
↑

Multiplication (Repeated addition)

↑
AND
↑

Conjunction (logical addition)

The '+' symbol
is read as
x OR y

$$\begin{array}{l} x + y \\ 0 + 0 = 0 \\ 0 + 1 = 1 \\ 1 + 0 = 1 \\ 1 + 1 = 1 \end{array}$$

Disjunction (Logical multiplication)

$x \cdot y$ The dot symbol
is read as x AND y

$$\begin{array}{l} x \cdot y \\ 0 \cdot 0 = 0 \\ 0 \cdot 1 = 0 \\ 1 \cdot 0 = 0 \\ 1 \cdot 1 = 1 \end{array}$$

1 and 0,
high and low,
true and false,
on and off.

George Boole, has written a book "An Investigation of the laws of thought". This book is based on logic and Probabilities.

Later on Claude E. Shannon at Massachusetts Institute of Technology wrote a thesis. Today Boolean Algebra is the back bone of Computer Circuit analysis.

Three basic laws of Boolean Algebra are same as normal algebra.

1. Commutative law:- $A + B = B + A$, $A \cdot B = B \cdot A$

2. Associative law: $(A + B) + C = A + (B + C)$, $A \cdot (B \cdot C) = (A \cdot B) \cdot C$

3. Distributive law: i) $A(B + C) = AB + AC$
ii) $A + B \cdot C = (A + B)(A + C)$

x = variables
 y = stores decimal
numbers in
normal algebra.
But in boolean
algebra it will
store only
binary values

algebra it will store only binary values

Logic Gates: A logic gate is defined as an electronic circuit with two or more input signals and have one output signal and it depicts the logic clearly.

There are 3 types of logic Gates :

1. Basic Gates

- OR gate
- AND gate
- NOT gate

2. Universal Gates

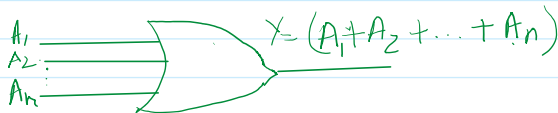
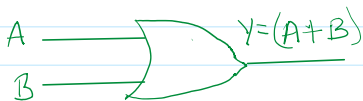
- NOR gate
- NAND gate

3. Exclusive Gates

- XOR gate
- XNOR gate

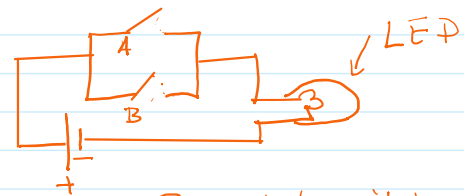
BASIC GATES

1. OR Gate :- An OR gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on, if any of the input signal is high or 1 or true or on. This is called (logic-1) gate.



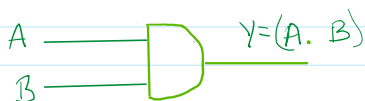
	A	B	Y
0	0	0	0
1	0	1	1
2	1	0	1
3	1	1	1

$2^n = 2^2 = 4$
combinations
(0, 1, 2, 3)



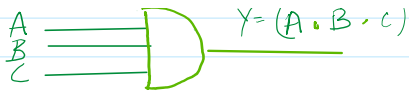
Parallel switch connection

ii) AND Gate :- An AND gate is a logic ckt. with '2' or more input signals and '1' output signal. The output signal will be high or 1 or true or on, if all of the input signal is high or 1 or true or on. The logic is represented by dot '.' symbol.

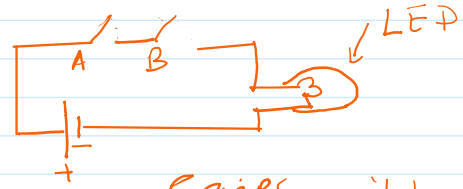
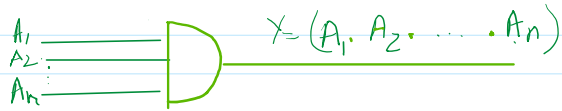


	A	B	Y
0	0	0	0
1	0	1	0
2	1	0	0
3	1	1	1

$2^n = 2^2 = 4$
combinations
(0, 1, 2, 3)

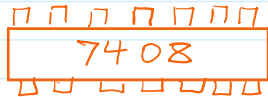


1	0	0	0
2	1	0	0
3	1	1	1

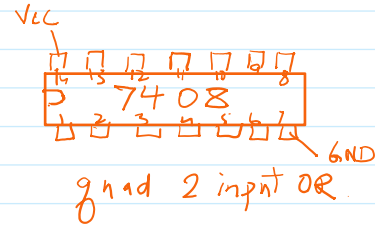


Series switch connection

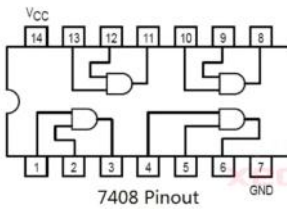
Integrated ckt. (IC)



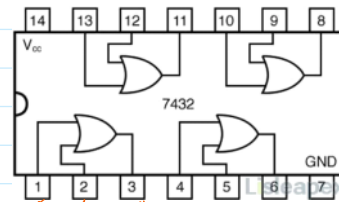
Quad 2 input AND



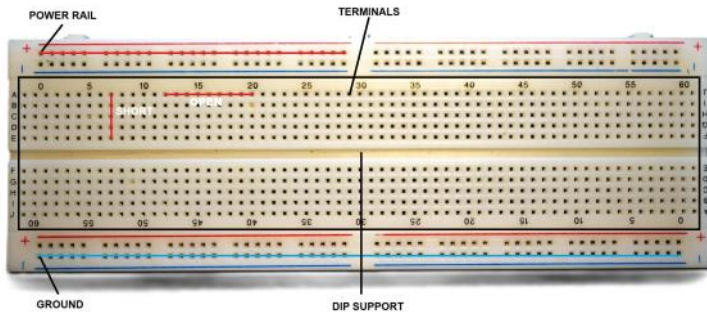
Quad 2 input OR



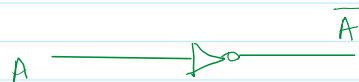
7408 Pinout



input output

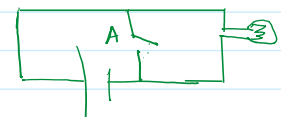


NOT Gate

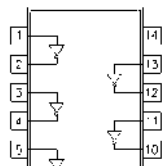


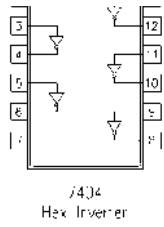
A	$\bar{A}$
0	1
1	0

A inverter or NOT gate is a ckt. that takes one input signal and output signal is always opposite or complement of the input signal.



Inverter ckt.
OR
short ckt





Inverter ckt.
or
short ckt.

NOT GATE IC